Introduction

Weighted kappa extends Cohen's kappa to ordinal data by crediting partial agreement: near-miss disagreements (mild vs moderate) weigh less than dramatic disagreements (mild vs severe). It is the standard inter-rater agreement statistic for Likert-type scales, radiology grading, and symptom-severity ratings.

Prerequisites

Cohen's kappa; ordinal measurement.

Theory

$$\kappa_w = 1 - \frac{\sum_{ij} w_{ij} f_{ij}}{\sum_{ij} w_{ij} e_{ij}},$$

where w_{ij} is the disagreement weight between category i and j, f_{ij} observed cell frequency, e_{ij} expected under chance.

Weight schemes: - **Linear** $w_{ij} = |i-j|/(k-1)$ for k categories. - **Quadratic** $w_{ij} = (i-j)^2/(k-1)^2$ – more forgiving of near-miss disagreements.

Quadratic is most common for multi-category ordinal scales.

Assumptions

Category ordering is meaningful and equally spaced; two raters; independent ratings.

R Implementation

```r
library(psych)

set.seed(2026)
n <- 100
# Two raters on a 5-point ordinal scale
rater1 <- sample(1:5, n, replace = TRUE,
                 prob = c(0.1, 0.2, 0.4, 0.2, 0.1))
# Rater 2 agrees within +-1 with prob 0.8, otherwise random
rater2 <- ifelse(rbinom(n, 1, 0.8) == 1,
                 pmax(1, pmin(5, rater1 + sample(-1:1, n, replace = TRUE))),
                 sample(1:5, n, replace = TRUE))

cohen.kappa(cbind(rater1, rater2))$kappa    # unweighted
cohen.kappa(cbind(rater1, rater2),
            w = "squared")$weighted.kappa
cohen.kappa(cbind(rater1, rater2),
            w = "linear")$weighted.kappa
```

Output & Results

Unweighted kappa (~0.35), linear-weighted (~0.55), quadratic-weighted (~0.70); quadratic weights reward near-miss agreement more heavily.

Interpretation

"Weighted kappa with quadratic weights was 0.72 (95 % CI 0.62-0.82), consistent with substantial agreement; unweighted kappa of 0.35 understates agreement by not crediting near-miss ratings."

Practical Tips

- Use quadratic weights for most clinical ordinal scales; they reflect that 1-step disagreements matter far less than 3-step.
- Linear weights are appropriate when category spacing is more uniform-linear.
- Always specify the weight scheme when reporting weighted kappa.
- For continuous data with measurement error, use the intraclass correlation coefficient (ICC) instead.
- Quadratic-weighted kappa equals ICC(3, 1) under certain assumptions – the two methods converge for ordinal Likert scales.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale